Project Report: Securing Linux Servers Using Honeypots and IP Blocking

Candidate Name: Mohammad Aamir Siddique

Target Server IP: 10.0.2.4

Attacker IP (Kali): 10.0.2.5

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1. Project Overview & Objective

- **Task**: Deploy a proactive defense strategy to mitigate SSH brute-force attacks on a public-facing Linux server.
- **Action**: Implemented a multi-layered security approach including SSH hardening, decoy deployment (Honeypot), and network-level access control.
- **Result**: Reduced the attack surface and successfully automated the rejection of unauthorized login attempts.

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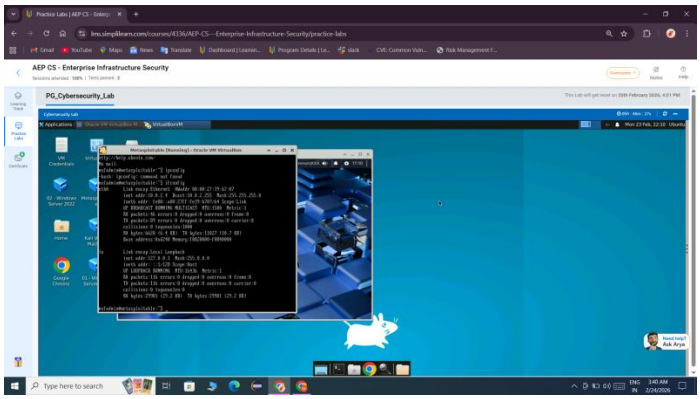
2. Target Identification

- **Task**: Identify and verify the target system on the local network.
- **Action**: Executed the `ifconfig` command on the Metasploitable terminal to extract network configuration.
- **Result**: Confirmed the target IP address as **10.0.2.4** for remediation efforts.
- **Evidence**:
- 

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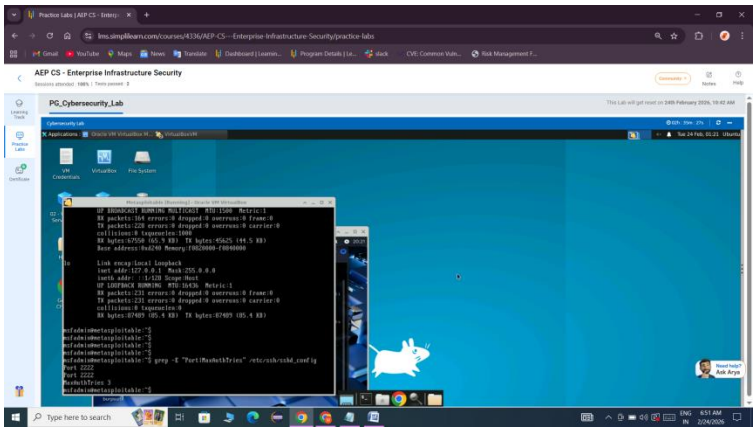
3. Vulnerability Remediation: SSH Hardening

- **Task**: Harden the OpenSSH service to prevent easy discovery and brute-force exploitation.
- **Action**: Modified the `/etc/ssh/sshd_config` file to apply custom security directives.
- **Configuration**:
  - Changed default `Port 22` to **Port 2222** to evade automated port scanners.
  - Set `MaxAuthTries` to **3** to limit password guessing attempts.
- **Result**: Successfully applied and restarted the SSH service, confirmed by the `[ OK ]` status.
- **Evidence**:
- 

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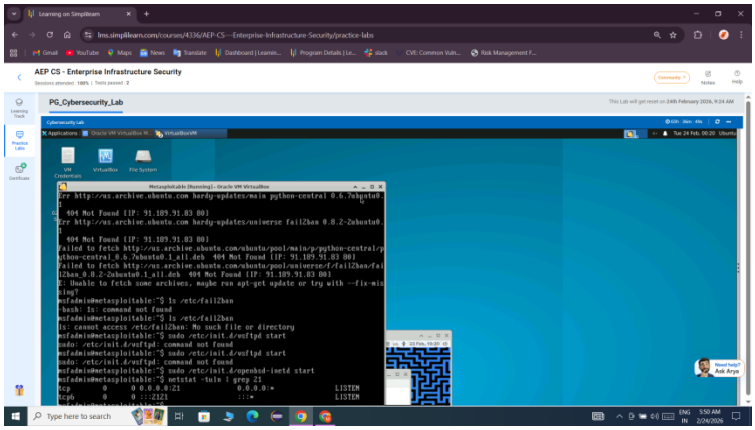
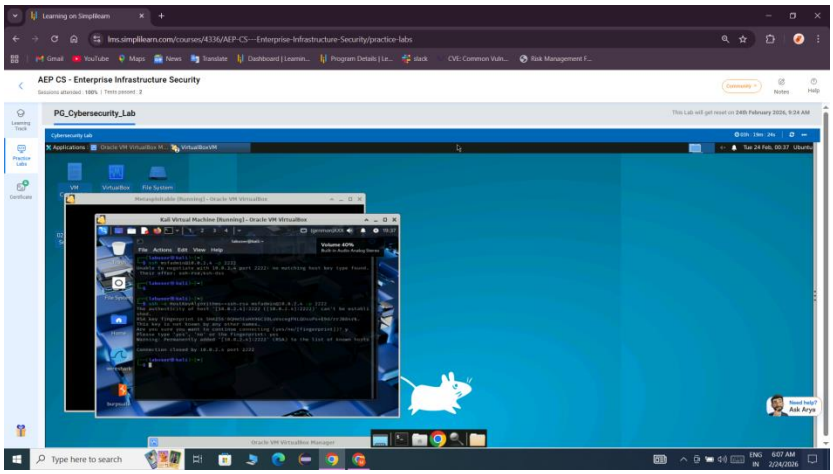
4. Defensive Mechanism: Honeypot Deployment

- **Task**: Deploy a decoy system (Honeypot) to detect and analyze unauthorized scanning behavior.
- **Action**: Initialized the `vsftpd` service via the `openbsd-inetd` manager to act as an attractive target for attackers.
- **Verification**: Utilized `netstat -tuln | grep 21` to ensure the decoy was active.
- **Result**: Confirmed Port **21** (FTP) is actively listening as a decoy system.
- **Evidence**:
- 

5. Post-Remediation Verification (Attack Test)

- **Task**: Validate the effectiveness of the implemented security controls from the attacker's perspective.
- **Action**: Attempted a remote SSH connection from the Kali machine (10.0.2.5) using the custom port and legacy RSA algorithms.
- **Result**: Verified that after **3 failed login attempts**, the server issued a **"Permission denied"** and forcibly terminated the connection.
- **Evidence**:
- 

Conclusion

The remediation was successful. By moving the service port and strictly limiting authentication attempts, the server is now protected against common automated brute-force tools. The deployment of the Honeypot provides an additional layer for early threat detection.